

README

Imbalanced and Balanced Disaster Tweet Classification

Project Overview

This project implements and compares Logistic Regression, Random Forest and Neural Network models on an imbalanced text dataset (Disaster Tweets). The performance is evaluated before and after applying imbalance handling techniques.

1. Dataset

- Dataset file used: tweets.csv
Dataset Setup
 - Download dataset from Kaggle:
<https://www.kaggle.com/datasets/vstepanenko/disaster-tweets/data>
 - Place the file in the project root directory

2. Technologies Used

- Python
 - pandas, numpy
 - scikit-learn
 - imbalanced-learn
 - nltk(natural language tool kit)
 - matplotlib
 - PyTorch (torch,torch vision)

Installation and Setup Instructions

Follow the steps below to set up the project environment.

Step 1: Installation

- Install Python
- Verify installation: `python --version`
- Download and install Visual Studio Code
- Install the following extensions:
 - Python (by Microsoft)
 - Jupyter (by Microsoft)

Step 2: Install Required Libraries

- Run the following command: `pip install pandas numpy scikit-learn matplotlib seaborn nltk imbalanced-learn torch notebook`

Step 3: Execution Instructions

-Run using Jupyter Notebook (VS Code)

1. Open the project folder in VS Code

2. Open the file: AI_Project_Group_7.ipynb

3. Click Run All Cells

-Make sure dataset(tweets.csv) is in the same folder as the notebook

-Run cells in order (top to bottom)

1. Models Implemented

- Logistic Regression
 - Random Forest
 - Neural Network

2. Imbalance Handling Techniques

- Class Weighting
 - Random Oversampling
 - SMOTE

3. Evaluation Metrics

- Accuracy
 - Precision
 - Recall
 - F1-score
 - ROC-AUC
 - Precision-Recall Curve
 - Confusion Matrix

4. Output

- Model comparison before and after imbalance handling
 - Performance metrics printed in notebook
 - Graphs for ROC and Precision-Recall curves

5. File Structure

- project_folder/
 - |— AI_Project_Group_7.ipynb
 - |— tweets.csv
 - |— README.pdf
 - |— report.pdf
 - |— Requirements.txt

6. Author / contributors

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